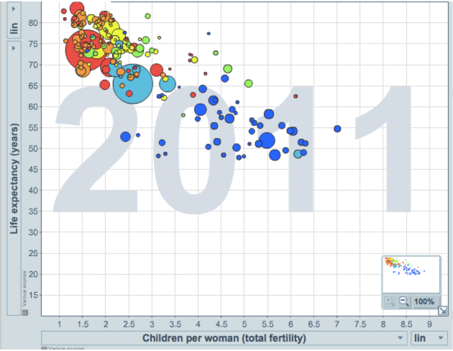
Numerical Data & Continuous Distributions

Numerical data

Scatterplots

Scatterplots are useful for visualizing the relationship between two numerical variables.

Do life expectancy and total fertility appear to be associated or independent? Was the relationship the same throughout the years, or did it change?



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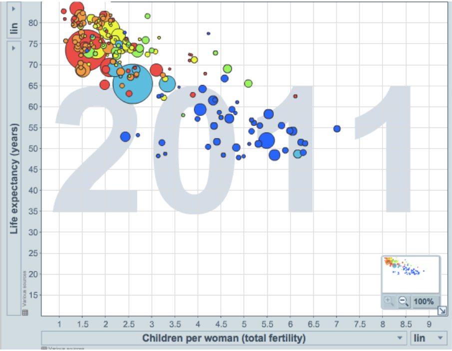
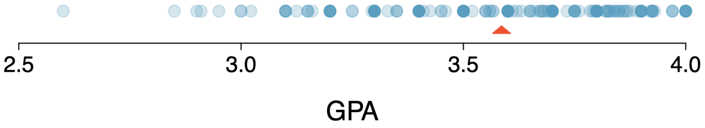


They appear to be linearly and negatively associated — as fertility increases, life expectancy decreases — and this relationship changed over time.

Dot plots & the mean

Dot plots are useful for visualizing one numerical variable. The **mean** (average) — marked with a triangle below — is one way to measure the center of a distribution.

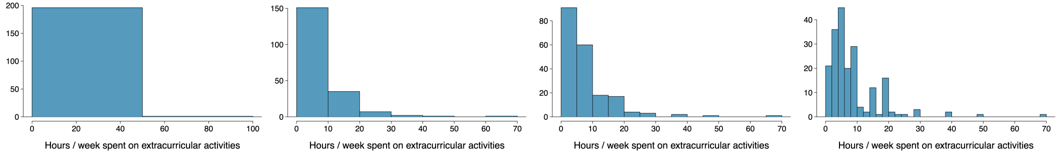


$$\bar{x} = \frac{x_1 + x_2 + \cdots + x_n}{n}$$

The sample mean $\bar{x}$ is a **point estimate** of the population mean μ — not perfect, but usually good if the sample is representative.

Histograms & bin width

Histograms show the **density** of the data — where values are relatively more common — and are especially useful for describing **shape**. The chosen **bin width** can change the story the histogram tells.

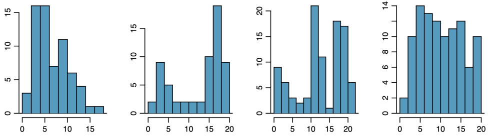


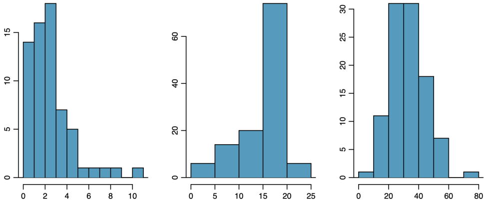
Which of these are useful? Which reveal too much? Which hide too much?

Shape of a distribution

Modality: does the histogram have one prominent peak (unimodal), several (bi/multimodal), or none (uniform)?

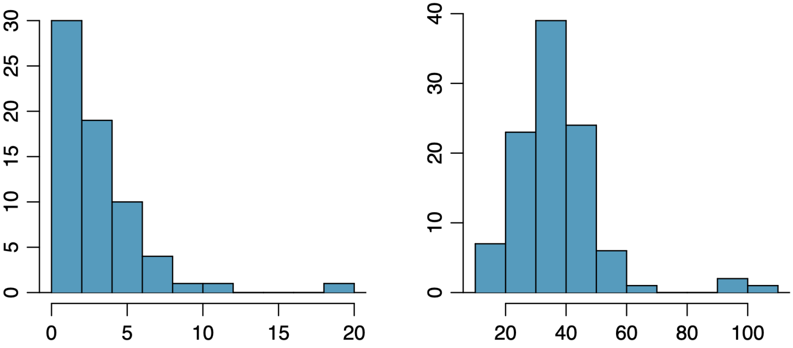
Skewness: is the histogram right skewed, left skewed, or symmetric? (Named for the side of the long tail.)





Unusual observations

Are there any unusual observations, or potential **outliers**?



Outliers are worth flagging: they can signal extreme skew, data-entry errors, or genuinely interesting features of the data.

Variance and standard deviation

Variance is roughly the average squared deviation from the mean:

$$s^2 = \frac{\sum_{i=1}^{n}(x_i - \bar{x})^2}{n-1}$$

The **standard deviation** is the square root of the variance, in the same units as the data:

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For a sample of students' sleep hours, $\bar{x} = 6.71$ and $n = 217$:

$$s^2 = \frac{(5-6.71)^2 + (9-6.71)^2 + \cdots + (7-6.71)^2}{217-1} = 4.11 \Rightarrow s = \sqrt{4.11} \approx 2.03 \text{ hours}$$

We square the deviations so that observations equally far from the mean count equally, and larger deviations are weighted more heavily.

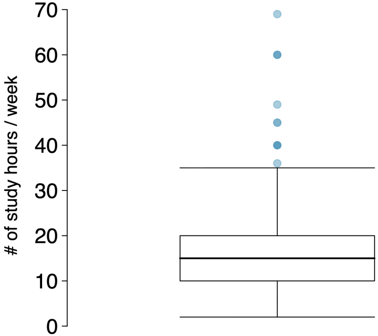
Median, quartiles, and IQR

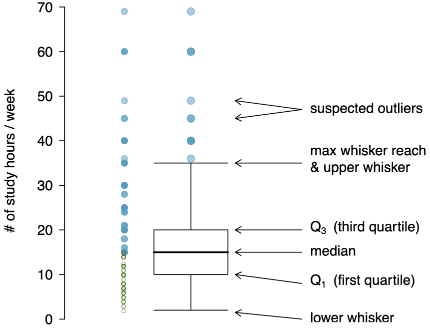
- The **median** splits the ordered data in half — 50% of values fall below it, so it's also the 50^{th} percentile.
  - Odd n: the middle value. Even n: the average of the two middle values.
- The 25^{th} percentile is the first quartile, **Q1**. The 75^{th} percentile is the third quartile, **Q3**.
- The middle 50% of the data spans the **interquartile range**:

$$IQR = Q3 - Q1$$

Box plots

The box represents the middle 50% of the data; the thick line is the median.





Whiskers extend up to $1.5 \times IQR$ beyond the quartiles:

$$\text{max whisker reach} = Q3 + 1.5 \times IQR \quad \text{or} \quad Q1 - 1.5 \times IQR$$

A point beyond that reach is a potential **outlier**.

Robust statistics

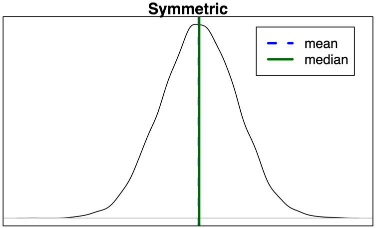
Median and IQR are more **robust** to skew and outliers than mean and SD.

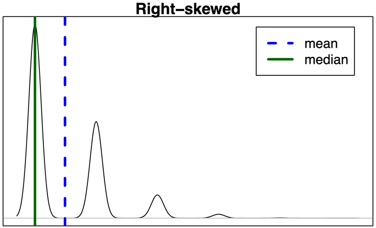
scenario	median	IQR	$\bar{x}$	s
original data	190K	200K	245K	226K
move largest to \$10M	190K	200K	309K	853K
move smallest to \$10M	200K	200K	316K	854K

For **skewed** distributions, prefer median/IQR to describe center/spread. For **symmetric** distributions, mean/SD are typically more informative.

Mean vs. median

- **Symmetric:** mean $\approx$ median
- **Right-skewed:** mean $>$ median
- **Left-skewed:** mean $<$ median





If you wanted to estimate the typical household income for a student, would you be more interested in the mean or the median?

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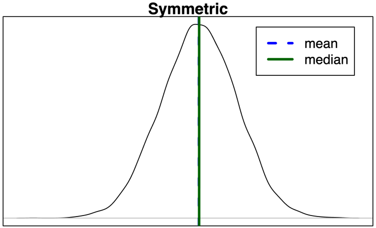

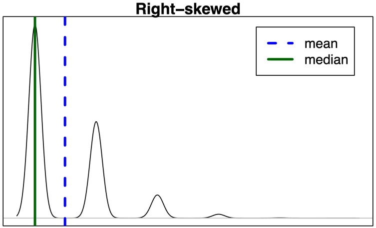

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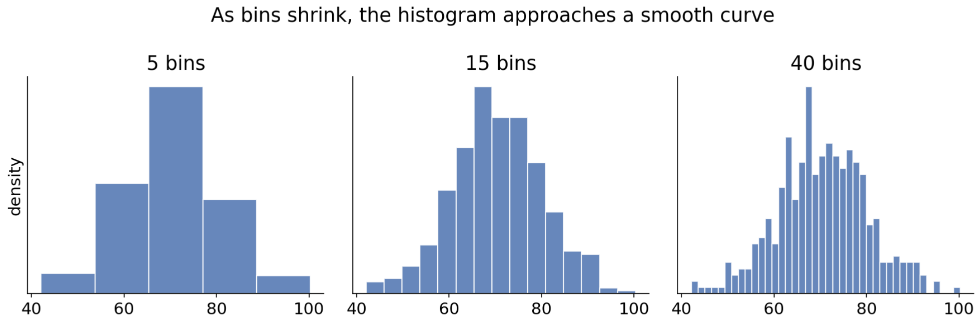
Median — it's not pulled around by a few very high (or low) incomes.

Continuous distributions

From histograms to density curves

So far, every distribution we've looked at has been a histogram — data grouped into bins. As the sample size grows and the bins shrink, the histogram's shape approaches a smooth curve called a **density curve**.

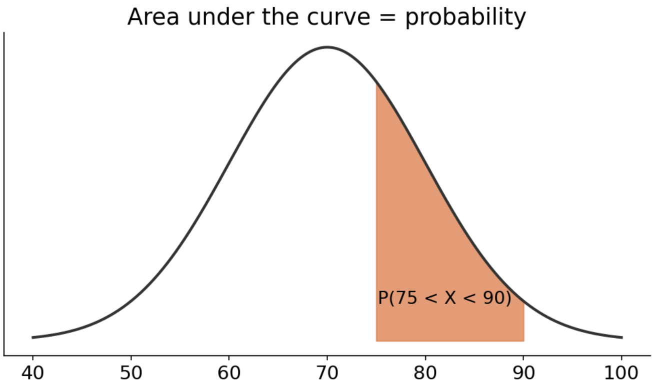


A density curve is how we describe the distribution of a **continuous** random variable — one that can take any real value, not just discrete counts.

Density curves: area = probability

For a continuous random variable, we can't ask "what's the probability of exactly this value?" (that probability is essentially zero). Instead:

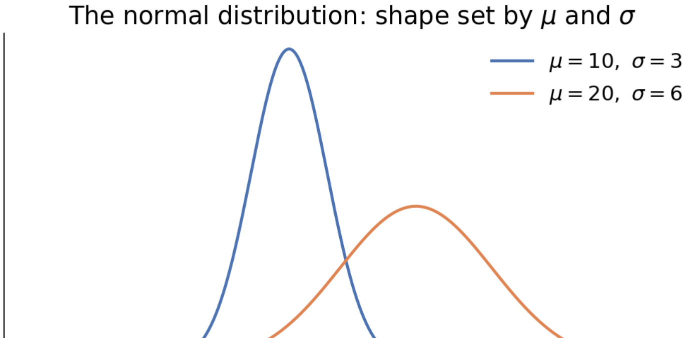
- The **total area** under a density curve equals 1.
- The probability that X falls in some range is the **area under the curve** over that range.



The normal distribution

The **normal distribution** is the most important continuous distribution — many real-world variables (heights, test scores, measurement errors) are approximately normal.

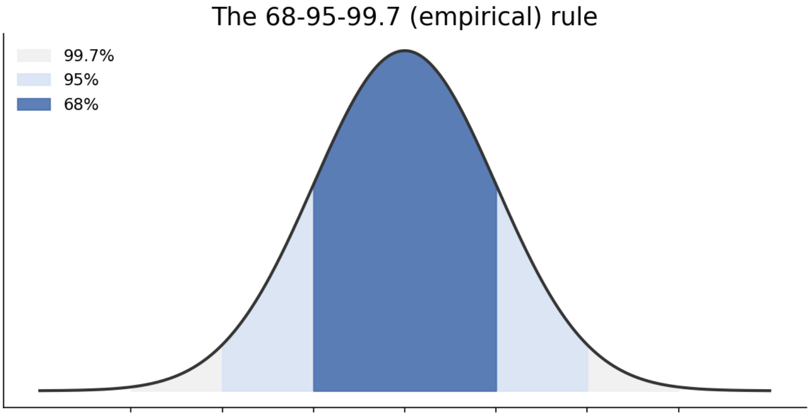
- Symmetric, bell-shaped, single peak.
- Fully described by two parameters: the mean μ (center) and standard deviation σ (spread).



The 68-95-99.7 rule

For any normal distribution:

- About **68%** of values fall within 1 SD of the mean.
- About **95%** fall within 2 SD.
- About **99.7%** fall within 3 SD.



Using the normal model

Suppose SAT math scores are approximately normal with $\mu = 500$ *and* $\sigma = 100$*. What percent of students score between 400 and 600?*

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What percent score above 700?

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400 and 600 are exactly 1 SD below and above the mean — so by the 68-95-99.7 rule, about **68%** of students fall in that range.

> *What percent score above 700?*

700 is 2 SD above the mean. About 95% fall within $\pm 2\sigma$, leaving 5% split between the two tails — so about **2.5%** score above 700.

Tying it together

- Monday, we described **categorical** data (bar plots, proportions) and built **discrete** probability distributions from countable outcomes.
- Today, we described **numerical** data (histograms, mean, SD, shape) and saw that as data become dense and continuous, that same shape becomes a **density curve**.
- The normal distribution lets us go from "here's what our sample's histogram looks like" to "here's a probability model we can use to answer questions about values we haven't observed."

Friday: we'll build these same summaries and plots in code.